

Equations

Week 3C

Overview

Formulas and Absolute Value Equations

Key Ideas - Working with Formulas

- Changing the Subject of an Equation*
- Introduction to Absolute Value Equations*
- Solving Absolute Value Equations*

Key Idea – Working with Formulas

In this section, we'll focus on solving equations that involve more than one variable. The process is similar to solving both linear and nonlinear equations. Before we begin, it's important to define what a formula is and explain what the subject of a formula means.

A formula is a type of equation which represent a special relationship between the variables. A formula shows how different quantities are related to each other. A formula will involve more than one variable.

The subject of a formula is the single variable that is being worked out. It can be recognised as the letter on its own on one side of the equals sign

For example, the formula $E = mc^2$ represents the relationship between energy (E), mass (m) and the speed of light squared (c^2) and is called “the theory of relativity”. The **subject** of the formula is E .

A formula can be used to **evaluate the subject** when values of all other variables are known. To find the value of ‘ E ’ in the example above, you substitute values for ‘ m ’ and ‘ c ’ in the right-hand side of the equation, then evaluate.

Example:

Use the formula $E = mc^2$ to calculate the energy (E), measured in Joules (J), released from a blazing campfire if 4 large logs, weighing a total of $5kg$ (m), are placed in the fire and the speed of light (c) is approximately $3 \times 10^8 m/s$.

Substitute the known values into the formula $E = mc^2$:

$$\begin{aligned} E &= 5 \times (3 \times 10^8)^2 \\ &= 4.5 \times 10^{17} J \end{aligned}$$

Example:

The volume of a sphere with radius r units is given by the formula $V = \frac{4}{3}\pi r^3$. Find the volume of a sphere with a radius of 4.15 cm. Write your answer correct to the nearest cm^3 .

Substitute the known values into the formula $V = \frac{4}{3}\pi r^3$:

$$\begin{aligned} V &= \frac{4}{3} \times \pi \times 4.15^3 \\ V &= 299 cm^3 \end{aligned}$$

Key Idea – Changing the Subject of a Formula

We can rearrange a formula so that any variable is the subject. To change the subject of a formula we apply **inverse operations** in the same way as we do when solving an equation, remembering to keep the **formula balanced**. That is, what you do to isolate the variable, must be applied to both sides of the equation.

Specifically,

- Add/subtract a number or expression to/from **both sides**.
- Multiply/divide **both sides** by a number or an expression.
- Square/square root of **both sides**.
- Cube/cube root of **both sides**.

***Note:** If given values for any variables, it may be easier to firstly substitute the known values, into the formula, then solve for the unknown variable.*

Example:

Use the formula $E = mc^2$ to calculate the mass (m), measured in kilograms (kg) released from a blazing campfire of 4 large logs. Given the speed of light (c) is approximately $3 \times 10^8 \text{ m/s}$ and the energy emitted in Joules (J) of $4.5 \times 10^{17} J$.

Substitute the known values into the formula $E = mc^2$:

$$4.5 \times 10^{17} = m \times (3 \times 10^8)^2$$

$$m = \frac{4.5 \times 10^{17}}{(3 \times 10^8)^2}$$

$$m = 5 \text{ kg}$$

Example:

The volume V of a sphere with radius r units is given by the formula $V = \frac{4}{3}\pi r^3$. Find the radius of a sphere whose volume is 300cm^3 . Write your answer in centimetres, correct to 3 significant figures.

Step 1: Substitute known values into the formula $V = \frac{4}{3}\pi r^3$:

$$300 = \frac{4}{3}\pi r^3$$

Step 2: Solve for r by making r the subject:

$$300 = \frac{4}{3}\pi r^3$$

$$\frac{4}{3}\pi r^3 = 300 \quad \text{Swap sides so that } r^3 \text{ is on the left}$$

$$r^3 = \frac{3 \times 300}{4\pi} \quad \text{Divide both sides by } \frac{4}{3} \text{ and } \pi$$

$$r = \sqrt[3]{\frac{3 \times 300}{4\pi}} \quad \text{Take the cube root of both sides}$$

$$r = 4.15\text{cm}$$

Example:

Rearrange the formula for the kinetic energy, $E_k = \frac{1}{2}mv^2$, to make velocity (v) the subject.

$$E_k = \frac{1}{2}mv^2$$

$$\frac{1}{2}mv^2 = E_k$$
 Swap sides so that v is on the left

$$mv^2 = 2E_k$$
 Multiply both sides by 2, to remove $\frac{1}{2}$

$$v^2 = \frac{2E_k}{m}$$
 Divide both sides by m

$$v = \pm \sqrt{\frac{2E_k}{m}}$$
 Take the square root of both sides, but velocity in this equation can never be negative

$$v = + \sqrt{\frac{2E_k}{m}}$$
 So we only take the positive answer

Example:

Make r the subject of this formula.

$$2B\sqrt{\frac{r}{y}} - 4 = a$$

$$2B\sqrt{\frac{r}{y}} = a + 4$$

Add 4 to both sides

$$\sqrt{\frac{r}{y}} = \frac{a + 4}{2B}$$

Divide both sides by $2B$

$$\frac{r}{y} = \left(\frac{a + 4}{2B}\right)^2$$

Square both sides

$$r = y\left(\frac{a + 4}{2B}\right)^2$$

Multiply both sides by y

Exercise: Using the formula $S = ut + at$, evaluate u when $S = 144$, $a = 16$, $t = 3$.

$$S = ut + at$$

$$144 = u \times 3 + 16 \times 3$$

$$144 = 3u + 48 \quad (-48)$$

$$96 = 3u \quad (\text{swap})$$

$$3u = 96 \quad (\div 3)$$

$$u = 32$$

Exercise: Rearrange $y = 8 - \frac{x}{4}$ to make x the subject.

$$4 \times y = 4 \left(8 - \frac{x}{4} \right)$$

$$4y = 4 \times 8 - \frac{4 \times x}{4}$$

$$4y = 32 - x$$

$$(+x)$$

$$x + 4y = 32$$

$$(-4y)$$

$$x = 32 - 4y$$

Exercise: The formulas for total resistance of two resistors in parallel is given by the formula

$\frac{1}{R_{total}} = \frac{1}{R_1} + \frac{1}{R_2}$. Rearrange the formula to make total resistance the subject.

$$\frac{1}{R_{total}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{R_{total}} = \frac{R_2 + R_1}{R_1 R_2}$$

$$R_{total} \left(\frac{1}{R_{total}} \right) = \left(\frac{R_1 + R_2}{R_1 R_2} \right) \times R_{total}$$

$$1 = \frac{(R_1 + R_2) R_{total}}{R_1 R_2} \quad (\times R_1 R_2)$$

$$\rightarrow R_1 R_2 = (R_1 + R_2) R_{total} \quad (\text{swap})$$

$$R_{total} (R_1 + R_2) = R_1 R_2 \quad (\div (R_1 + R_2))$$

$$R_{total} = \frac{R_1 R_2}{R_1 + R_2}$$

Key Idea – Introduction to Absolute Value Equations

The size of a real number x is called its **modulus** or **absolute value** and is denoted by $|x|$.

We define absolute value of x , or modulus of x , by:

$$|x| = \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0 \end{cases}$$

Geometrically $|x|$ is the distance of the point representing x on the number line from the origin, regardless of the direction of the point from zero .

Similarly, $|x - a|$ is the distance between x and a on the number line and $|a - b|$ is the distance between a and b .

Note: *When finding the modulus of a real number disregard the sign.*

Example: Find $|5|$.

Using the number line:

The distance of $|5|$ from the origin is 5 units. Hence $|5| = 5$



Using the absolute value definition:

$$|x| = \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0 \end{cases} \quad \text{we can see } |5| = \begin{cases} 5, & \text{as } 5 \geq 0 \\ -x, & \text{if } x < 0 \end{cases}$$

$$\text{Hence } |5| = 5$$

Example: Find $|-5|$

Using the number line:

The distance of $|-5|$ from the origin is 5 units. Hence $|-5| = 5$



Using the absolute value definition:

$$|x| = \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0 \end{cases} \quad \text{we can see } |-5| = \begin{cases} x, & \text{if } x \geq 0 \\ -(-5), & \text{as } -5 < 0 \end{cases}$$

$$\text{Hence } |-5| = 5$$

Example: Find $|2 + -7|$

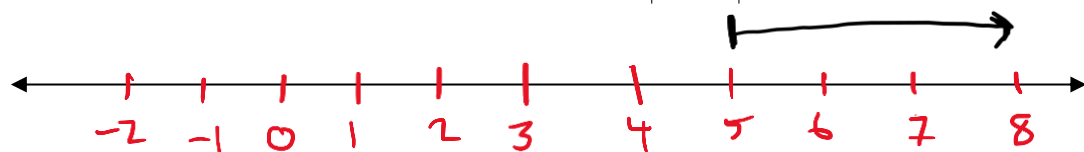
$|2 + -7| = |2 - 7|$ is the distance between 2 and 7.

$$\text{Hence } |2 - 7| = |-5| = 5$$



Exercises:

- a) Use the number line to evaluate $|5-8|$



$|5-8|$ is the distance between 5 & 8

$$|5-8| = 3$$

- b) Evaluate each of the following using the absolute value definition $|x| = \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0 \end{cases}$

$$|-2| = -(-2) = 2$$

$$|-10-13| = |-23| = -(-23) = 23$$

Properties of Absolute Values

Property	Example
$ x \geq 0$	$ 20 \geq 0, -20 \geq 0$
$ x = -x $	$ 6 = 6, \text{ and } -6 = 6$
$ ab = a \times b $	$ 3 \times -7 = 3 \times -7 = 21$
$\left \frac{a}{b}\right = \frac{ a }{ b }$	$\left \frac{63}{-9}\right = \frac{ 63 }{ -9 } = 7$
$a^2 = a ^2$	$(-9)^2 = -9 ^2 = 81$
$\sqrt{a^2} = a $	$\sqrt{(-9)^2} = -9 = 9$

Note: The properties $(\sqrt{a})^2 = a$ and $\sqrt{a^2} = |a|$ will be used in later topics.

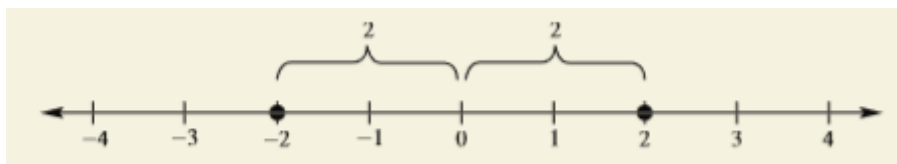
Key Idea – Solving Absolute Value Equations

In order to solve absolute value equations, we will first look at two examples using the number line.

Example: Solve $|x| = 2$

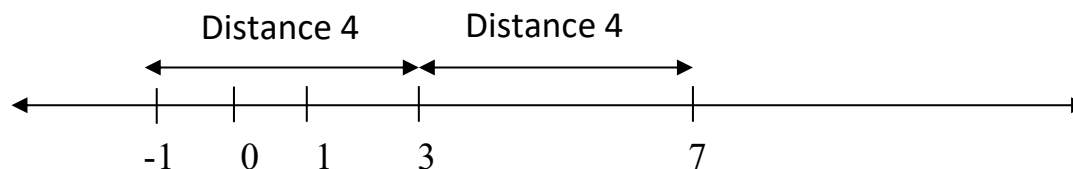
$|x| = 2$ means the distance of x from zero is 2 units (in either direction).

Therefore, $x = \pm 2$



Example: Solve $|x - 3| = 4$.

$|x - 3| = 4$ means the distance between the real number x and 3 is 4 units (in either direction).



Therefore, $x = -1, 7$

Equations involving absolute value can also be solved algebraic.

The steps for the algebraic solution are:

- Step 1:** Set the expression inside the absolute value notation equal to + and - the quantity on the other side of the equation.
- Step 2:** Solve for the unknown in each equation.
- Step 3:** Check your answer analytically by substituting the solutions into the original equation. If the $LHS = RHS$ then the solution exists, if $LHS \neq RHS$ then the solution does not exist.

Example: Solve $|x-3|=4$

Step 1: $(x-3) = \pm 4$

Step 2: $(x-3) = +4$ or $(x-3) = -4$
 $x = 7$ $x = -1$

Step 3: *Check solutions by substitution*

If $x = 7$: $LHS = |x-3|$
 $= |7-3| = |4|$
 $= 4$

$RHS = 4$

$LHS = RHS$

If $x = -1$: $LHS = |x-3|$
 $= |-1-3| = |-4|$
 $= 4$

$RHS = 4$

$LHS = RHS$

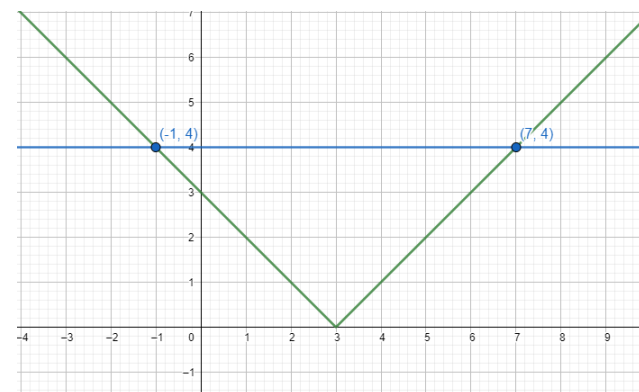
$\therefore x = 7$ and $x = -1$ are both solutions.

Graphical Solution

Graphically the solutions of

$|x-3|=4$ are the x coordinates of the points

of intersection of the graphs of $y = |x-3|$ and $y = 4$.



Exercise: Use algebra to solve the absolute value equation $|2x - 3| = 8$

$$2x - 3 = \pm 8$$

$$2x - 3 = 8$$

$$2x = 11$$

$$x = \frac{11}{2}$$

$$2x - 3 = -8$$

$$2x = -5$$

$$x = -\frac{5}{2}$$

Check

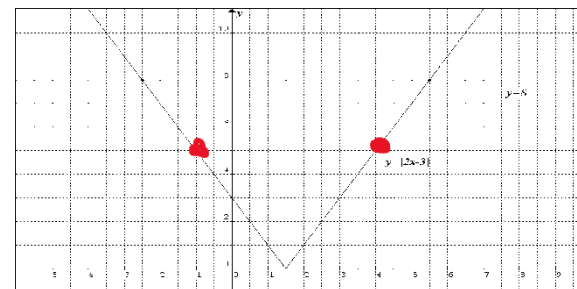
for $x = \frac{11}{2}$

$$\begin{aligned} \text{LHS} &= |2x - 3| \\ &= |2 \times \frac{11}{2} - 3| \\ &= |8| \\ &= 8 \\ &= \text{RHS} \end{aligned}$$

for $x = -\frac{5}{2}$

$$\begin{aligned} \text{LHS} &= |2x - 3| \\ &= |2 \times (-\frac{5}{2}) - 3| \\ &= |-8| \\ &= 8 \\ &= \text{RHS} \end{aligned}$$

Both $x = \frac{11}{2}$ & $x = -\frac{5}{2}$
are solutions



Example: Solve $|2x| = 3x + 5$

Step 1: $(2x) = \pm(3x + 5)$

Step 2: $2x = 3x + 5$ or $(2x) = -(3x + 5)$
 $-x = 5$ $2x = -3x - 5$
 $x = -5$ $5x = -5$
 $x = -1$

Step 3: *Check solution analytically:*

$$\begin{aligned} \text{If } x = -5: \quad LHS &= |2x| \\ &= |2 \times (-5)| \\ &= |-10| \\ &= 10 \\ RHS &= 3x + 5 \\ &= 3 \times (-5) + 5 \\ &= -10 \\ LHS &\neq RHS \end{aligned}$$

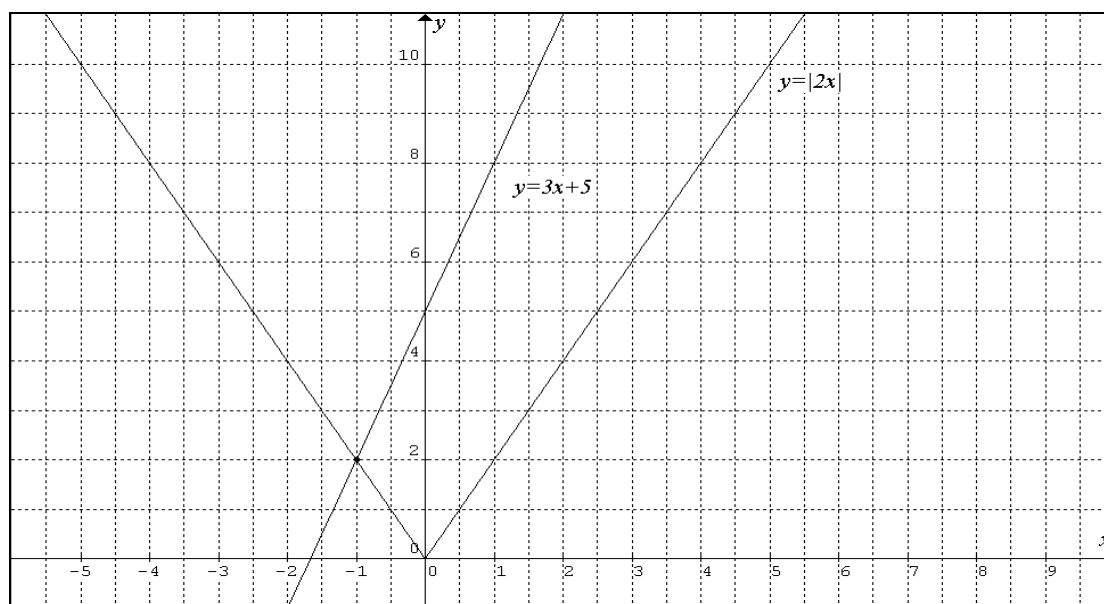
$\therefore x = -5$ is not a solution

$$\begin{aligned} \text{If } x = -1: \quad LHS &= |2x| \\ &= |2 \times (-1)| \\ &= |-2| \\ &= 2 \\ RHS &= 3x + 5 \\ &= 3 \times (-1) + 5 \\ &= 2 \\ LHS &= RHS \end{aligned}$$

$\therefore x = -1$ is a solution

Graphical Solution

Graphically the solution of $|2x| = 3x + 5$ is the x coordinate of the point of intersection of the graphs of $y = |2x|$ and $y = 3x + 5$.



Exercise: Solve the equation $|x-2| = 3(2x+1)$ algebraically

$$x-2 = \pm 3(2x+1)$$

$$x-2 = 3(2x+1)$$

$$x-2 = 6x+3$$

$$-5 = 5x$$

$$\boxed{x = -1}$$

$$\text{LHS} = |x-2|$$

$$= |-1-2|$$

$$= |-3|$$

$$= 3$$

$$\text{RHS} = 3(2x+1)$$

$$= 3(2 \times (-1) + 1)$$

$$= -1$$

$$\text{LHS} \neq \text{RHS}$$

$$x-2 = -3(2x+1)$$

$$x-2 = -6x-3$$

$$7x = -1$$

$$\boxed{x = -\frac{1}{7}}$$

$$\text{LHS} = |x-2|$$

$$= |-\frac{1}{7}-2|$$

$$= |-\frac{15}{7}|$$

$$= \frac{15}{7}$$

$$\text{RHS} = 3(2x+1)$$

$$= 3(2 \times (-\frac{1}{7}) + 1)$$

$$= \frac{15}{7}$$

$$= \text{LHS}$$

check

Only $x = -\frac{1}{7}$ is
the solution

***Note:** When we have an absolute value expression on each side of an equation, it is easier to fix the sign of one of them and vary the sign of the other.*

Example: Solve $|3x+1| = |2x+4|$

Step 1: $3x+1 = \pm(2x+4)$

Step 2:

$$\begin{array}{lcl} 3x+1 = 2x+4 & \text{or} & 3x+1 = -(2x+4) \\ x = 3 & & 3x+1 = -2x-4 \\ & & 5x = -5 \\ & & x = -1 \end{array}$$

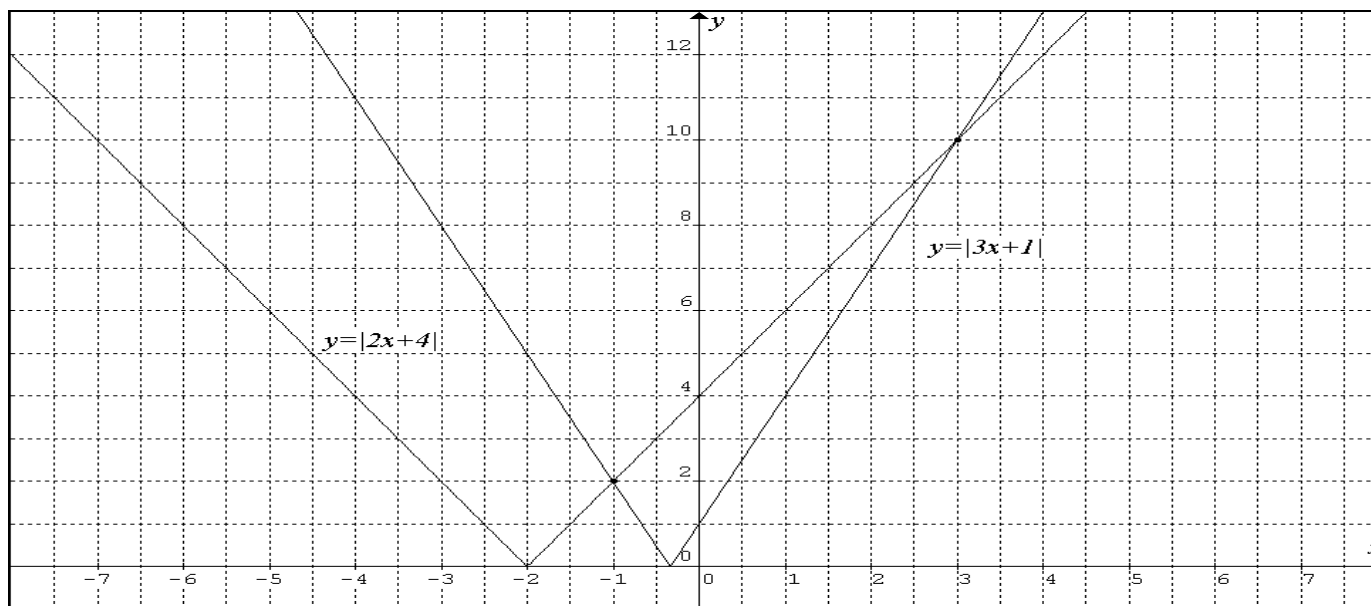
Step 3: *Check solution analytically:*

	$LHS = 3(3)+1 $		$LHS =$
if $x = 3$	$= 10$	if $x = -1$	$= 3(-1)+1 $
			$= 2$
	$RHS = 2(3)+4 $		$RHS = 2(-1)+4 $
	$= 10$		$= 2$
	$LHS = RHS$		$LHS = RHS$

$\therefore x = 3$ and $x = -1$ are both solutions.

Graphical Solution

Drawn below are the graphs of $y = |3x + 1|$ and $y = |2x + 4|$



The points of intersection of these graphs indicate that the solutions to the equation are $x = -1$ and $x = 3$

Exercise: Solve the equation $|3x+1| = |2x+4|$ algebraically.

$$3x+1 = \pm(2x+4)$$

$$3x+1 = 2x+4$$

$$\boxed{x=3}$$

$$3x+1 = -(2x+4)$$

$$3x+1 = -2x-4$$

$$5x = -5$$

$$\boxed{x=-1}$$

Check

$$\begin{aligned}\text{LHS} &= |3x+1| \\ &= |3 \times 3 + 1| \\ &= |10| \\ &= 10\end{aligned}$$

$$\begin{aligned}\text{RHS} &= |2x+4| \\ &= |2 \times 3 + 4| \\ &= |10| \\ &= 10 \\ &= \text{LHS}\end{aligned}$$

$$\begin{aligned}\text{LHS} &= |3x+1| \\ &= |3 \times (-1) + 1| \\ &= |-2| \\ &= 2 \\ \text{RHS} &= |2x+4| \\ &= |2 \times (-1) + 4| \\ &= |2| \\ &= 2 \\ &= \text{LHS}\end{aligned}$$

Both $x=3$ & $x=-1$
are solutions.